

Rethinking the Relationship between the Power Law and Hierarchical Structures

Kai Nakaishi^{1,*} Ryo Yoshida² Kohei Kajikawa^{3,*} Koji Hukuahima² Yohei Oseki²

¹RIKEN, Japan

²The University of Tokyo, Japan

³National Institute for Japanese Language and Linguistics, JAPAN

kai.nakaishi@riken.jp

{yoshiryo0617, k-hukushima, oseki}@g.ecc.u-tokyo.ac.jp

kohei-kajikawa@ninjal.ac.jp

Abstract

Statistical analysis of corpora provides an approach to quantitatively investigate natural languages. This approach has revealed that several power laws consistently emerge across different corpora and languages, suggesting the universal principles underlying languages. Particularly, the power-law decay of correlation has been interpreted as evidence for underlying hierarchical structures in syntax, semantics, and discourse. This perspective has also been extended to child languages and animal signals. However, the argument supporting this interpretation has not been empirically tested. To address this problem, this study examines the validity of the argument for syntactic structures. Specifically, we test whether the statistical properties of parse trees align with the implicit assumptions in the argument. Using English corpora, we analyze the mutual information, deviations from probabilistic context-free grammars (PCFGs), and other properties in parse trees, as well as in the PCFG that approximates these trees. Our results indicate that the assumptions do not hold for syntactic structures and that it is difficult to apply the proposed argument to child languages and animal signals, highlighting the need to reconsider the relationship between the power law and hierarchical structures.

Zipf’s law (Zipf, 2016) and Heaps’ law (Heaps, 1978), that consistently appear across different corpora and languages. These findings suggest the existence of universal principles of languages.

This study focuses on the power-law decay of correlation. In languages, distant elements can remain strongly correlated. For example, subject–verb agreement in English can occur at arbitrarily large distances, and a single sentence can have semantic influence over an entire text. Previous studies have measured correlations between words, characters, phones, and other elements using metrics such as correlation functions and mutual information (MI). These studies have consistently found that the correlation decays according to a power-law function of the distance between elements (Li, 1989a; Ebeling and Pöschel, 1994; Ebeling and Neiman, 1995; Tanaka-Ishii and Bunde, 2016; Lin and Tegmark, 2017; Takahashi and Tanaka-Ishii, 2017; Shen, 2019; Takahashi and Tanaka-Ishii, 2019; Sainburg et al., 2019; Mikhaylovskiy and Churilov, 2023; Nakaishi et al., 2024). This phenomenon indicates that the correlation is significant even at large distances.

Moreover, the power-law decay of correlation has been interpreted as evidence supporting the linguistic view that languages are organized hierarchically in syntax, semantics, and discourse (Alvarez-Lacalle et al., 2006; Altmann et al., 2012; Lin and Tegmark, 2017). This perspective has also been extended beyond adult human languages. Previous research has observed similar power-law behaviors in child languages and in animal signals, arguing that these may also possess underlying hierarchical structures (Sainburg et al., 2019; Sainburg and Gentner, 2021; Sainburg et al., 2022; Howard-Spink et al., 2024; Youngblood, 2024).

This interpretation is justified by the argument of Lin and Tegmark (2017). They showed that

1 Introduction

Natural languages exhibit characteristic statistical properties. Quantifying these properties through corpus-based analyses is crucial for understanding natural languages, providing insights into the mechanisms that give rise to the properties. Previous research has extensively explored this approach, uncovering various power laws, such as

*This work was conducted while KN and KK were at the University of Tokyo.

the decay of correlation follows a power law in sequences generated by probabilistic context-free grammars (PCFGs; Jelinek et al., 1992), which are a simple model for generating hierarchical tree structures. To be more precise, they proved that in PCFGs, the correlation decays exponentially in structures and assumed that the sequential distance grows exponentially with the structural distance. These give rise to the power-law decay of correlation in sequences.

However, this argument rests on several implicit assumptions. Previous studies have not tested their validity. To address this, we examine whether these assumptions hold for hierarchical structures in syntax, i.e., parse trees. Although parsing schemes vary in detail, syntactic structures have been well formalized, resulting in the availability of large-scale treebanks. This enables us to perform statistical analyses on treebanks to test these assumptions.

Specifically, we formalize three key questions essential for examining the validity of the assumptions: (i) Does the correlation decay exponentially in structures?; (ii) Does the sequential distance grow exponentially with the structural distance?; (iii) Are PCFGs adequate models for the hierarchical structures of natural languages? To answer these questions, we analyze treebanks.

As a result, our findings are negative for all three questions, indicating that the assumptions in the proposed argument do not hold for syntactic structures. Therefore, it is needed to reconsider the relationship between the power law and hierarchical structures.

In addition, we perform the same analyses on trees generated by the PCFG that approximates syntactic structures, to compare them with natural languages. These analyses highlight the differences between PCFGs and natural languages. Furthermore, their statistical properties are also different from those assumed in the argument by Lin and Tegmark (2017). This means that the PCFG does not converge to its asymptotic behavior within the scale of a single sentence. Consequently, it is difficult to apply the proposed argument not only to single sentences, but also to other types of short sequences such as child languages and non-human signals.

To investigate if the argument holds beyond the sentence level, future research will need to formalize the description of hierarchical structures of en-

tire texts, such as discourse. Otherwise, we need to seek for an alternative argument that account for the power-law behavior.

2 Background

2.1 Power-law Decay of Correlation

The significance of the power-law decay of correlation is clear in comparison with the exponential decay. For example, in signals generated sequentially by a Markov chain, the correlation C is asymptotically exponential with the distance r , i.e., $C \sim \exp(-\lambda r)$ (Li, 1990; Lin and Tegmark, 2017). In this case, if the distance is sufficiently larger than $\xi = 1/\lambda$, the correlation is negligible. Thus, ξ , which is called the *correlation length* in statistical physics (Stanley, 1971), means the scale within which the correlation is significant.

In contrast, the power-law decay, $C \sim r^{-\alpha}$, is slower than the exponential one for any correlation length ξ , meaning that the correlation is significant for any long distances. In other words, the scale of correlation diverges. Moreover, power-law functions are scale-invariant. For the scale transformation $r' = kr$, we can rescale the correlation by $C' = k^{-\alpha}C$ so that the relation remains the same, $C' \sim r'^{-\alpha}$. This property also implies the divergence of the scale.

Therefore, the power-law decay observed in various corpora means that the scale diverges in natural languages. Intuitively, a single element change can significantly influence the overall structure of a text.

2.2 Power Law and Hierarchical Structures

In the study of human languages and animal signals, the power-law decay of correlation has been interpreted as evidence of underlying hierarchical structures. This interpretation is based on the work of Lin and Tegmark (2017), which shows that the decay of correlation follows a power law in sequences generated by PCFGs. PCFGs generate tree structures from the root to leaves iteratively, where the state of a parent node determines the states of its children. The leaves correspond to a sequence, while the whole tree is the underlying hierarchical structure. PCFGs are a probabilistic extension of context-free grammars (CFGs; Chomsky, 1956). CFGs were originally introduced to describe the nested structures, such as the *if...then* construction. Due to their simplicity and ability to capture fundamental aspects

of hierarchical syntactic structures, PCFGs have been used to model syntactic structures (Charniak, 1997). Beyond linguistics, PCFGs have also been applied to represent hierarchical structures in diverse domains (Ellis et al., 2015; Worth and Stepney, 2005; Gilbert and Conklin, 2007; Knudsen and Hein, 1999; Harlow et al., 2012; Li, 1989b, 1991; Tano et al., 2020; Worth and Stepney, 2021).

We give an overview of the discussion by Lin and Tegmark (2017). To understand their discussion, it is useful to distinguish between *sequential* and *structural* distances, which we refer to as r_{seq} and r_{str} , respectively. The former is defined along the sequence. On the other hand, the latter is the distance in the hierarchical structure behind the sequence, as shown in Figure 1.

Lin and Tegmark (2017) have proven that the correlation asymptotically decays according to an exponential function of the structural distance, $C \sim \exp(-\lambda r_{\text{str}})$. They also assumed that generated trees are balanced, i.e., they have neither left- nor right-branching bias, as in Figure 1 (a). Letting the depth of the two nodes from the common ancestor be d , the structural distance is roughly $r_{\text{str}} \sim 2d$, and the sequential distance is asymptotically exponential with d , $r_{\text{seq}} \sim q^d$, where q is the typical number of children of each node. Therefore, the sequential distance grows exponentially with the structural distance, $r_{\text{seq}} \sim \exp(\mu r_{\text{str}})$, where $\mu = (\ln q)/2$. Combining $C \sim \exp(-\lambda r_{\text{str}})$ and $r_{\text{seq}} \sim \exp(\mu r_{\text{str}})$ leads to the power-law decay of correlation in sequences, $C \sim r_{\text{seq}}^{-\alpha}$, with $\alpha = \lambda/\mu$.

This discussion demonstrates that the correlation in sequences decays according to a power law if the sequences have hierarchical structures that satisfy the two properties: First, the correlation decays exponentially with the structural distance, $C \sim \exp(-\lambda r_{\text{str}})$; second, the structures are

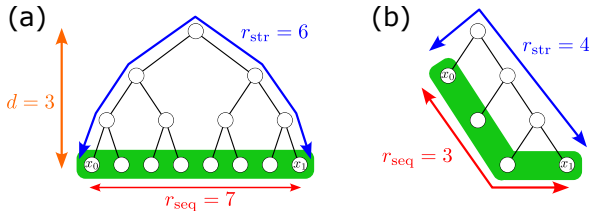


Figure 1: Sequential and structural distances. (a) If trees are balanced, the sequential distance grows exponentially with the structural distance, $r_{\text{seq}} \sim \exp(\mu r_{\text{str}})$. (b) If trees are strongly biased, the growth is slower. For example, the relation is linear, $r_{\text{seq}} \sim r_{\text{str}}$.

balanced, such that the growth of sequential distance is exponential with the structural distance, $r_{\text{seq}} \sim \exp(\mu r_{\text{str}})$.

It is widely accepted among linguists that the syntactic structure of a sentence is hierarchical. Moreover, formal descriptions of these structures have been well established. Therefore, this study aims to examine the validity of the proposed argument, focusing on syntactic structures.

Meanwhile, the proposed argument is not limited to syntactic structures in principle. It may also apply to other types of structures, such as semantic and discourse ones. For example, entire texts composed of many sentences intuitively appear to have hierarchical structures defined by paragraphs, sections, chapters, and so on. This hierarchical organization may explain why the decay of correlation follows a power law at scales beyond the sentence level, extending across entire texts. The structures may also be more abstract. Previous research has observed the power-law decay in sequences without syntax, such as child languages and animal signals, arguing that these also have underlying hierarchical structures (Sainburg et al., 2019; Sainburg and Gentner, 2021; Sainburg et al., 2022; Howard-Spink et al., 2024; Youngblood, 2024).

However, the existence and the formalization of these types of hierarchical structures remain unclear. Therefore, we are currently unable to assess whether the proposed argument accounts for the power law at scales beyond the sentence level or in early-stage or non-human languages.

2.3 Rethinking Previous Research

The argument by Lin and Tegmark (2017) raises several important questions that must be addressed to determine whether it can apply to syntactic structures. First, do hierarchical structures in natural languages actually exhibit the behaviors assumed in their argument? More specifically:

- i) Does the correlation in hierarchical structures decay exponentially with the structural distance?
- ii) Are the hierarchical structures sufficiently balanced such that the sequential distance grows exponentially with the structural distance?

Particularly, the latter appears to contradict the well-known fact that syntactic structures typically exhibit a left- or right-branching bias (Hawkins, 1990; Dryer, 1992) (for instance, see Figure 2). In

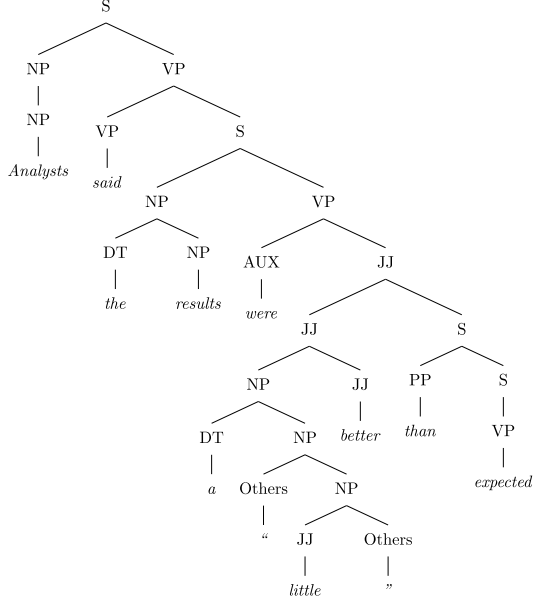


Figure 2: Example of a parse tree from the BLLIP corpus, preprocessed as described in Section 3.1. This tree has a strong right-branching bias.

such cases, the growth is slower than exponential. For example, Figure 1 (b) illustrates a case where the growth is linear, $r_{\text{seq}} \sim r_{\text{str}}$.

Moreover, it is unlikely that PCFGs model syntactic structures sufficiently well. Linguistically, some languages cannot be described by CFGs (Shieber, 1985; Culy, 1985; Senuma and Aizawa, 2024). In addition, empirically, naive parsers based on PCFGs are insufficient for parsing (Johnson et al., 2006; O’Donnell et al., 2009). If syntactic structures deviate from PCFGs significantly, the discussion by Lin and Tegmark (2017) based on PCFGs should not be applied to them. This leads to the last question:

- iii) Do PCFGs describe the statistical properties of hierarchical structures sufficiently?

In this study, we perform statistical analyses on parse trees to address these questions.

3 Methods

3.1 Datasets and Preprocessing

We adopt the Brown Laboratory for Linguistic Information Processing 1987-89 Corpus Release 1 (BLLIP; Charniak et al., 2000), a large-scale English treebank that provides Penn Treebank (Marcus et al., 1993)-style syntactic annotations for three years’ worth of Wall Street Journal articles. This treebank contains approximately 30 million

words and one million sentences in total. In advance, we remove the phonological null elements such as traces of movement and ellipsis to ensure the alignment with the WikiText-103 corpus (Merity et al., 2017) analyzed in Appendix D, which does not include traces.

The number of child nodes differs across nodes in the dataset, making it difficult to align nodes across different parse trees. To address this issue, we binarize the trees using the `Tree.chomsky_normal_form` method from the NLTK library (Bird and Loper, 2004). We then apply the `Tree.collapse_unary` method from the same library to remove the unary branches produced by the former method. In addition, because a larger number of possible node states requires more data for reliable statistical estimation, we replace phrasal and part-of-speech (POS) tags with a smaller set of categories. Specifically, we refer to the Universal Dependencies (de Marneffe et al., 2021), which contains fewer types of tags than the Penn Treebank-style corpus, and group tags that tend to have similar types of dependents and heads. We present the eleven categories used in our analyses in Appendix A. An example of a preprocessed parse tree is shown in Figure 2.

3.2 Mutual Information

As a metric for correlations, we use MI as in Lin and Tegmark (2017). The MI between two tags is defined by

$$I = \sum_{x_0, x_1} P(x_0, x_1) \ln \frac{P(x_0, x_1)}{P(x_0)P(x_1)}, \quad (1)$$

where $P(x_0)$ and $P(x_1)$ are the marginal probabilities of the first and second tags, and $P(x_0, x_1)$ is their joint probability. A naive estimate of I can be obtained by substituting the empirical distributions $\hat{P}(x_0)$, $\hat{P}(x_1)$, and $\hat{P}(x_0, x_1)$ into Eq. (1). However, this estimation is biased positively due to finite samples (Li, 1990). To reduce this bias, we decompose the MI into

$$I = S_0 + S_1 - S_{01}, \quad (2)$$

where S_0 , S_1 , and S_{01} are the Shannon entropy, defined by $S_0 = -\sum_{x_0} P(x_0) \ln P(x_0)$, $S_1 = -\sum_{x_1} P(x_1) \ln P(x_1)$, and $S_{01} = -\sum_{x_0, x_1} P(x_0, x_1) \ln P(x_0, x_1)$. Then we estimate the entropy S_0 using the method proposed

by Grassberger (2003):

$$\hat{S}_0 = \psi(N) - \frac{1}{N} \sum_{x_0} n_{x_0} \psi(n_{x_0}), \quad (3)$$

where ψ is the digamma function, n_{x_0} is the number of samples such that the former tag is x_0 , and $N = \sum_{x_0} n_{x_0}$ is the total number of samples. S_1 and S_{01} are estimated in a similar manner. By computing $\hat{S}_0$, $\hat{S}_1$, and $\hat{S}_{01}$ from the empirical distributions and using Eqs. (2) and (3), we can estimate the MI with a smaller bias.

To evaluate the sufficiency of the sampling, we compute the MI estimate while varying the number of samples, N_{data} . If the estimate converges as N_{data} increases, this suggests that sampling bias is negligible.

To estimate MI in sequences, we sample N_{data} pairs of POS nodes separated by a sequential distance r_{seq} , from the dataset. For MI in structures, we sample N_{data} pairs of nodes at a fixed structural distance r_{str} , where the nodes are either POS or phrasal. We treat ordered pairs as distinct to avoid arbitrarily selecting one of the two. For example, we count pairs (AUX, PP) and (PP, AUX), corresponding to *were* and *than*, in Figure 2 as distinct to each other if both pairs are sampled.

3.3 Growth of Sequential Distance

To address question (ii), we investigate how the sequential distance depends on the structural distance. Specifically, for each r_{seq} and r_{str} , we count the number of POS-tagged node pairs over the whole treebank whose sequential and structural distances are r_{seq} and r_{str} , respectively. We also compute the average of sequential distance for each structural distance r_{str} .

3.4 Context-free Independence Breaking

To examine the validity of PCFGs as models of syntactic structures, it is necessary to quantify the extent to which structures deviate from PCFGs. We employ a metric for this quantification, called *context-free independence breaking* (CFIB; Nakaishi and Hukushima, 2024).

The essential property of PCFGs, which we refer to as the *context-free independence*, is that the distribution of states of each node depends only on its parent. Consequently, given the states of two nodes, their children are mutually independent.

Therefore, we can quantify the deviation from PCFGs by the MI between two pairs under the

condition that the states of their parents are fixed, as presented in Figure 3. Specifically, provided that the states of two nodes are x_0 and x_1 , the MI between their children is defined as

$$J_{x_0 x_1} = \sum_{y_0, z_0, y_1, z_1} P(y_0, z_0, y_1, z_1 | x_0, x_1) \times \ln \frac{P(y_0, z_0, y_1, z_1 | x_0, x_1)}{P(y_0, z_0 | x_0, x_1) P(y_1, z_1 | x_0, x_1)}. \quad (4)$$

Here, $P(y_0, z_0 | x_0, x_1)$ denotes the conditional probability that the left and right children of the node with label x_1 are y_0 and z_0 , respectively. Similarly, $P(y_1, z_1 | x_0, x_1)$ refers to those for the node with label x_1 . $P(y_0, z_0, y_1, z_1 | x_0, x_1)$ is the conditional joint probability of both pairs of children. By construction, this metric is non-negative for any tree structures and zero for PCFGs.

For the estimation of the CFIB, we sample N_{data} pairs of nodes that are tagged as x_0 and x_1 , with the structural distance of r_{str} . Then, we compute the empirical distributions of tags on the children of the nodes. From these distributions, we estimate J in the same procedure explained in the previous section. We also distinguish between pairs and their permutations, as in the estimation of MI.

3.5 PCFG Approximation

We also construct a PCFG that approximates the distribution of parse trees. This allows for a more detailed comparison between syntactic structures and PCFGs. The PCFG is constructed using the maximum likelihood estimation, which is equivalent to simply counting the frequency of each production rule in the dataset. Using the PCFG derived from the preprocessed dataset from BLLIP, we generate trees 1.6×10^7 times, with up to 1×10^2 iterations per generation. This process resulted in 13,256,547 terminated trees. We then apply the same analyses to these generated trees as to parse trees.

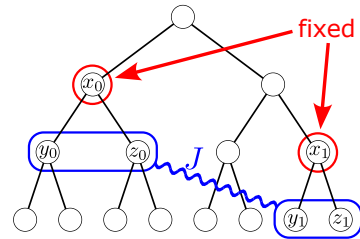


Figure 3: CFIB J is the MI between the children of two nodes whose states are fixed.

3.6 Fitting

We test whether the decay of correlation follows an exponential or power-law form by fitting the data with an exponential function, $A \exp(-\lambda r)$, and a power-law function, $B r^{-\alpha}$. To perform the fitting, we apply the least squares method after log-scaling the MI and CFIB values, because fitting with a linear scale puts too much weight on data points at shorter distances. We also fit the growth of sequential distance using an exponential function, $C \exp(\mu r)$, and a power-law function, $D r^\beta$, without log-scaling the values. For both fitting, we use the `lmfit` Python package (Newville et al., 2014). In the following results, we present the fitted parameters A , λ , and others along with the reduced chi-square statistic, χ^2_ν .

4 Results

4.1 Mutual Information in Sequences

First, we confirm that the correlation in sequences decays according to a power law. Figure 4 shows the MI I_{POS} between POS tags as a function of the sequential distance r_{seq} for different data sizes N_{data} . The estimated values converge as N_{data} increases, indicating that the estimation bias is sufficiently small. In addition, the decay is better fitted by a power-law function than by an exponential one. This result indicates that the correlation between POS tags within a sentence follows a power law, consistent with previous research. Similar results are also observed for the WikiText corpus, see Appendix D.

4.2 Mutual Information in Structures

In this subsection, we address question (i), whether the exponential decay of correlation with the structural distance holds in syntactic struc-

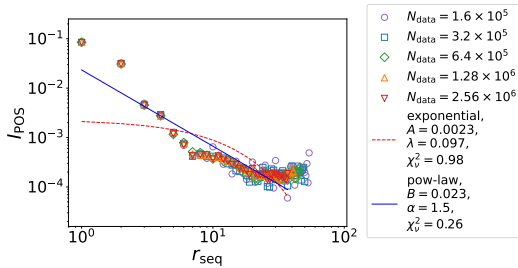


Figure 4: MI I_{POS} between POS tags as a function of the sequential distance r_{seq} . The fitted exponential and power-law decays for $N_{\text{data}} = 2.56 \times 10^6$ are also presented.

tures. The estimated MI between POS or phrasal tags as a function of the structural distance is shown in Figure 5. As in the previous subsection, the convergence of the estimated values with increasing N_{data} indicates that the bias is negligible. Interestingly, model fitting reveals that the correlation decays according to a power law rather than an exponential form with respect to the structural distance. This result indicates that syntactic structures do not meet the condition necessary for the previously proposed argument to hold. We have also confirmed that similar results are obtained for the unbinarized case and the WikiText corpus, as detailed in Appendices C and D.

4.3 Growth of Sequential Distance

The second question is (ii): Does the sequential distance grow exponentially with increasing structural distance? To examine this, we count the number of POS tag pairs such that the structural and sequential distances are r_{str} and r_{seq} , respectively. The results are shown in Figure 6. We also compute the averages sequential distance for each structural distance, represented by black circles, and fit the results using both exponential and power-law functions. The fitting reveals that the growth follows a power law. However, the trend becomes less clear for larger structural distances $r_{\text{str}} \gtrsim 50$ because of data sparsity. Notably, the power law exponent $\beta = 0.87$ is smaller than 1, indicating that the growth is sublinear, i.e., slower than linear. As demonstrated in Figure 1 (b), this slow growth quantitatively reflects a storing branching bias in syntactic structures. These findings clearly show that the second condition does not hold for syntactic structures. We have also observed similar results for the unbinarized case and for the WikiText corpus in Appendices C and D.

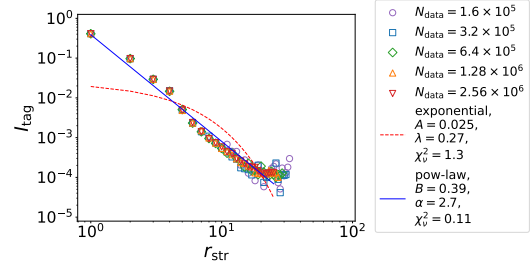


Figure 5: MI I_{tag} between POS or phrasal tags as a function of the structural distance r_{str} . The fitted exponential and power-law decays for $N_{\text{data}} = 2.56 \times 10^6$ are also presented.

4.4 Context-free Independence Breaking

Question (iii) asks whether PCFGs are valid models for the hierarchical structures of natural languages. In this section, we measure the CFIB J as the metric of deviations from PCFGs for syntactic structures. In Figure 7, we present the results for fixed nodes being $(x_0, x_1) = (\text{NP}, \text{NP})$, the most frequent pair in the corpus. The convergence of the estimates with increasing N_{data} confirms that the estimation bias is negligible. Moreover, the CFIB values are clearly positive, meaning that syntactic structures do not have the essential property of PCFGs. Furthermore, the metric decays according to a power law with respect to structural distance, suggesting that the context-free independence is significantly broken even at large distances. These findings lead to answering question (iii) in the negative. As shown in Appendix B, similar behaviors are observed for other frequent node pairs $(x_0, x_1) = (\text{NP}, \text{VP})$ and $(x_0, x_1) = (\text{VP}, \text{VP})$. In addition, Appendix D shows that the decay of J also follows a power law for the WikiText corpus.

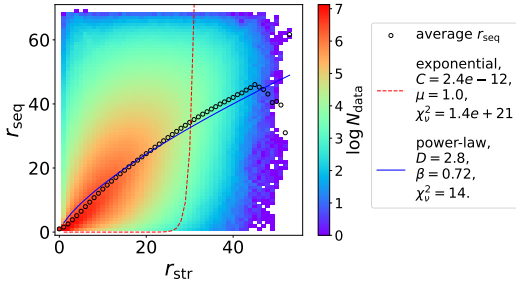


Figure 6: Number of POS tag pairs such that the structural and sequential distances are r_{str} and r_{seq} , respectively. Black circles represent the average sequential distance for each structural distance.

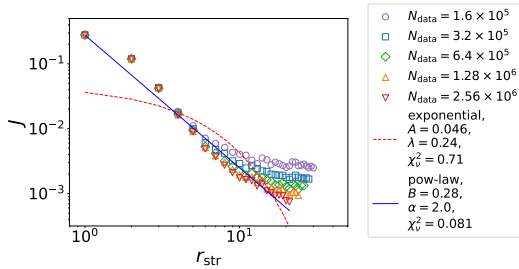


Figure 7: CFIB J for $(x_0, x_1) = (\text{NP}, \text{NP})$ as a function of the structural distance r_{str} . The fitted exponential and power-law decays for $N_{\text{data}} = 2.56 \times 10^6$ are also presented.

4.5 PCFG approximation

To deepen our understanding of the results above in comparison with PCFGs, we apply the same analyses to the PCFG constructed to approximate the BLLIP corpus. Figure 8 summarizes the results. The most striking difference between the parse trees and the PCFG appears in the CFIB metric, shown in Figure 8 (d). The estimated value of this metric does not saturate but continues to decrease even for large data sizes. This behavior is consistent with the fact that CFIB is identically zero for PCFGs by definition.

For the other statistical quantities, the differences are more moderate. As shown in Figure 8 (a), the MI in sequences decays according to a power law at short sequential distances, $r_{\text{seq}} \lesssim 10$, while for large sequential distances, $r_{\text{seq}} \gtrsim 10$, it increases substantially, in contrast to the trend in parse trees. The MI in structures, as shown in Figure 8 (b), exhibits a power-law decay as in parse trees. In Figure 8 (c), the average sequential distance grows more slowly than exponentially and even than linearly, indicating that the generated trees are strongly biased. While this behavior is qualitatively similar to that in syntactic structures, the estimated exponent $\beta = 0.45$ significantly differs from that for the parse trees, where $\beta = 0.87$.

Although these findings may suggest that the PCFG partially captures the statistical properties of parse trees, the mechanism underlying the power-law decay of correlation in sequences clearly differs from the one proposed in Lin and Tegmark (2017), in which the decay of MI in structures and the growth of sequential distance are exponential. Importantly, our finding that the MI in structures decays according to a power law in the PCFG suggests that the distance within a sentence is too short for the MI to converge to its asymptotic exponential decay.

This indicates that the proposed argument is difficult to apply to short sequences such as single sentences. In such cases, it becomes impossible to distinguish whether the process underlying the sequences is Markov-like or PCFG-like. This limitation also arises when analyzing child languages and animal languages, because available datasets do not contain sequences of sufficient length. Indeed, previous studies have analyzed sequences shorter than $\sim 10^2$ phones, syllables, words, or action elements (Sainburg et al., 2019; Sainburg

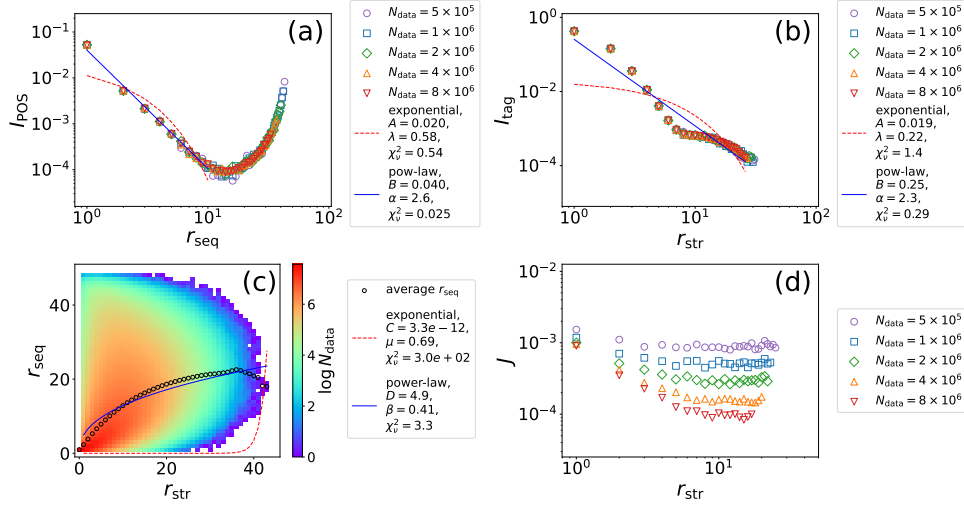


Figure 8: Statistical properties of trees generated by the PCFG obtained from the maximum likelihood estimation of syntactic structures. (a) MI I_{POS} between POS tags. (b) MI I_{tag} between POS or phrasal tags. (c) Distribution and average of sequential distances. (d) CFIB J for $(x_0, x_1) = (\text{NP}, \text{NP})$. Fitting was performed for $N_{\text{data}} = 8 \times 10^6$ in (a) and (b), and for $r_{\text{str}} \leq 10$ in (a).

and Gentner, 2021; Sainburg et al., 2022; Howard-Spink et al., 2024; Youngblood, 2024).

5 Discussion and Conclusion

Lin and Tegmark (2017) attributed the power-law decay of correlation in natural languages to underlying hierarchical structures, which has inspired numerous studies on both human languages and animal signals. However, their explanation raises several important questions, as discussed in Section 2.3. To address these, the present study analyzes the statistical properties of hierarchical syntactic structures in natural languages, to test the questions essential for the proposed argument.

The analyses answer all the questions negatively: (i) The correlation in syntactic structures decays according to a power law with respect to the structural distance; (ii) The growth of the sequential distance is slower than linear with the structural distance, due to the strong branching bias in syntactic structures; (iii) The statistical properties of syntactic structures differ from those of PCFGs, particularly in the CFIB. These results reveal that the argument by Lin and Tegmark (2017) cannot account for the mechanism underlying the power law at the sentence level.

We also analyze the PCFG that approximates the parse trees. The behavior of CFIB in the PCFG clearly differs from that in the parse trees. Furthermore, this analysis demonstrates that the asymptotic behavior of the PCFG does not emerge for

short sequences. This result highlights the difficulty of applying the proposed argument not only to single sentences in adult human languages but also to child languages and animal signals.

While further investigation is needed for other parsing schemes and corpora, we expect qualitatively similar results in more general settings. Indeed, we have confirmed similar results for unbinarized parse trees and for the WikiText corpus (Merity et al., 2017), see Appendices C and D for details. Whether these behaviors appear in other languages remains an open question. A key challenge is the limited availability of treebanks for languages other than English.

Another important open question is whether the argument by Lin and Tegmark (2017) applies at scales longer than sentences, that is, at the level of entire texts, where the statistical behaviors of PCFGs may be sufficiently close to their asymptotic behaviors. To extend our analysis to the level, a necessary first step is to formalize how to describe the hierarchical structures of entire texts and develop datasets annotated with such information.

Since the discussion in previous research does not account for the power law in natural languages, at least at the sentence level, alternative explanations are also needed. One promising direction is *critical phenomena* in statistical physics. These refer to the phenomena that statistical quantities exhibit power-law behaviors near *phase transition* point, where statistical properties of a system

change qualitatively. Indeed, as the temperature parameter of large language models varies, generated texts exhibit critical phenomena, sharing statistical properties with natural languages, including the power-law decay of correlation (Nakaishi et al., 2024).

Finally, more detailed characterizations of the statistical quantities are necessary. This study used simple fitting functions to distinguish between exponential and power-law behaviors. However, the observed statistical quantities exhibit more complex patterns. A closer examination of these patterns will be essential for understanding syntactic structures through statistical analysis.

Acknowledgments

This work was supported by JSPS KAKENHI Grant Nos. 23H01095, 23KJ0622, and 24H00087; JST, PRESTO Grant No. JPMJPR21C2; and the World-Leading Innovative Graduate Study Program for Advanced Basic Science Course at the University of Tokyo.

References

- Eduardo G Altmann, Giampaolo Cristadoro, and Mirko Degli Esposti. 2012. [On the origin of long-range correlations in texts](#). *Proceedings of the National Academy of Sciences*, 109(29):11582–11587.
- Enrique Alvarez-Lacalle, Beate Dorow, Jean-Pierre Eckmann, and Elisha Moses. 2006. [Hierarchical structures induce long-range dynamical correlations in written texts](#). *Proceedings of the National Academy of Sciences*, 103(21):7956–7961.
- Steven Bird and Edward Loper. 2004. [NLTK: The natural language toolkit](#). In *Proceedings of the ACL Interactive Poster and Demonstration Sessions*, pages 214–217, Barcelona, Spain. Association for Computational Linguistics.
- Eugene Charniak. 1997. [Statistical techniques for natural language parsing](#). *The AI Magazine*, 18:33–44.
- Eugene Charniak, Don Blaheta, Niyu Ge, Keith Hall, John Hale, and Mark Johnson. 2000. [BLLIP 1987-89 WSJCorpus Release 1 LDC2000T43](#). *Linguistic Data Consortium*.
- Noam Chomsky. 1956. [Three models for the description of language](#). *IRE Transactions on information theory*, 2(3):113–124.
- Christopher Culy. 1985. [The complexity of the vocabulary of bambara](#). *Linguistics and Philosophy*, 8:345.
- Matthew S. Dryer. 1992. [The greenbergian word order correlations](#). *Language*, 68(1):81–138.
- Werner Ebeling and Alexander Neiman. 1995. [Long-range correlations between letters and sentences in texts](#). *Physica A*, 215(3):233–241.
- Werner Ebeling and Thorsten Pöschel. 1994. [Entropy and Long-Range correlations in literary english](#). *Europhysics Letters*, 26(4):241–246.
- Kevin Ellis, Armando Solar-Lezama, and Josh Tenenbaum. 2015. [Unsupervised learning by program synthesis](#). In *Advances in Neural Information Processing Systems 28*.
- Édouard Gilbert and Darrell Conklin. 2007. [A probabilistic context-free grammar for melodic reduction ?](#) In *Proceedings of the International Workshop on Artificial Intelligence and Music, 20th International Joint Conference on Artificial Intelligence*.
- Peter Grassberger. 2003. [Entropy estimates from insufficient samplings](#). *arXiv preprint physics/0307138*.
- Daniel Harlow, Stephen H. Shenker, Douglas Stanford, and Leonard Susskind. 2012. [Tree-like structure of eternal inflation: A solvable model](#). *Physical Review D*, 85(6):063516.
- John A. Hawkins. 1990. [A parsing theory of word order universals](#). *Linguistic inquiry*, 21(2):223–261.
- Harold Stanley Heaps. 1978. *Information retrieval: Computational and theoretical aspects*. Academic Press, Inc.
- Matthew Honnibal, Ines Montani, Sofie Van Landeghem, and Adriane Boyd. 2020. [spaCy: Industrial-strength Natural Language Processing in Python](#).
- Elliot Howard-Spink, Misato Hayashi, Tetsuro Matsuzawa, Daniel Schofield, Thibaud Gruber, and Dora Biro. 2024. [Nonadjacent dependencies and sequential structure of chimpanzee](#)

- action during a natural tool-use task. *PeerJ*, 12:e18484.
- Filip Jelinek, John D. Lafferty, and Robert L. Mercer. 1992. [Basic methods of probabilistic context free grammars](#). In *Speech Recognition and Understanding*.
- Mark Johnson, Thomas L. Griffiths, and Sharon Goldwater. 2006. [Adaptor grammars: A framework for specifying compositional nonparametric bayesian models](#). In *Advances in Neural Information Processing Systems 19*.
- Nikita Kitaev, Steven Cao, and Dan Klein. 2019. [Multilingual constituency parsing with self-attention and pre-training](#). In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, pages 3499–3505, Florence, Italy. Association for Computational Linguistics.
- Nikita Kitaev and Dan Klein. 2018. [Constituency parsing with a self-attentive encoder](#). In *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 2676–2686, Melbourne, Australia. Association for Computational Linguistics.
- Bjarne Knudsen and Jotun Hein. 1999. [Rna secondary structure prediction using stochastic context-free grammars and evolutionary history](#). *Bioinformatics*, 15(6):446–454.
- Wentian Li. 1989a. [Mutual information functions of natural language texts](#). Technical Report 89-10-008, Santa Fe Institute.
- Wentian Li. 1989b. [Spatial 1/f spectra in open dynamical systems](#). *Europhysics Letters*, 10(5):395.
- Wentian Li. 1990. [Mutual information functions versus correlation functions](#). *Journal of statistical physics*, 60:823–837.
- Wentian Li. 1991. [Expansion-modification systems: a model for spatial 1/f spectra](#). *Physical Review A*, 43(10):5240.
- Henry W. Lin and Max Tegmark. 2017. [Critical behavior in physics and probabilistic formal languages](#). *Entropy*, 19(7):299.
- Mitchell P. Marcus, Beatrice Santorini, and Mary Ann Marcinkiewicz. 1993. [Building a large annotated corpus of English: The Penn Treebank](#). *Computational Linguistics*, 19(2):313–330.
- Marie-Catherine de Marneffe, Christopher D. Manning, Joakim Nivre, and Daniel Zeman. 2021. [Universal Dependencies](#). *Computational Linguistics*, 47(2):255–308.
- Stephen Merity, Caiming Xiong, James Bradbury, and Richard Socher. 2017. [Pointer sentinel mixture models](#). In *Internal Conference on Learning Representations*.
- Nikolay Mikhaylovskiy and Ilya Churilov. 2023. [Autocorrelations decay in texts and applicability limits of language models](#). *arXiv preprint arXiv:2305.06615*.
- Kai Nakaishi and Koji Hukushima. 2024. [Statistical properties of probabilistic context-sensitive grammars](#). *Physical Review Research*, 6(3):033216.
- Kai Nakaishi, Yoshihiko Nishikawa, and Koji Hukushima. 2024. [Critical phase transition in large language models](#). *arXiv preprint arXiv:2406.05335*.
- Matthew Newville, Till Stensitzki, Daniel B. Allen, and Antonino Ingargiola. 2014. [Lmfit: Non-linear least-square minimization and curve-fitting for python](#).
- Timothy J. O’Donnell, Joshua B. Tenenbaum, and Noah D. Goodman. 2009. [Fragment grammars: Exploring computation and reuse in language](#). Technical Report MIT-CSAIL-TR-2009-013, Massachusetts Institute of Technology.
- Tim Sainburg and Timothy Q. Gentner. 2021. [Toward a computational neuroethology of vocal communication: from bioacoustics to neurophysiology, emerging tools and future directions](#). *Frontiers in Behavioral Neuroscience*, 15:811737.
- Tim Sainburg, Anna Mai, and Timothy Q. Gentner. 2022. [Long-range sequential dependencies precede complex syntactic production in language acquisition](#). *Proceedings of the Royal Society B*, 289(1970):20212657.

Tim Sainburg, Brad Theilman, Marvin Thielk, and Timothy Q. Gentner. 2019. [Parallels in the sequential organization of birdsong and human speech](#). *Nature Communications*, 10(1):3636.

Hajime Senuma and Akiko Aizawa. 2024. [Computational complexity of natural morphology revisited](#). *Transactions of the Association for Computational Linguistics*, 12.

Huitao Shen. 2019. [Mutual information scaling and expressive power of sequence models](#). *arXiv preprint arXiv:1905.04271*.

Stuart M. Shieber. 1985. [Evidence against the context-freeness of natural language](#). *Linguistics and Philosophy*.

Harry E. Stanley. 1971. *Introduction to phase transitions and critical phenomena*. Oxford University Press.

Shuntaro Takahashi and Kumiko Tanaka-Ishii. 2017. [Do neural nets learn statistical laws behind natural language?](#) *PloS one*, 12(12):e0189326.

Shuntaro Takahashi and Kumiko Tanaka-Ishii. 2019. [Evaluating computational language models with scaling properties of natural language](#). *Computational Linguistics*, 45(3):481–513.

Kumiko Tanaka-Ishii and Armin Bunde. 2016. [Long-range memory in literary texts: On the universal clustering of the rare words](#). *PLoS One*, 11(11):e0164658.

Pablo Tano, Sergio Romano, Mariano Sigman, Alejo Salles, and Santiago Figueira. 2020. [Towards a more flexible language of thought: Bayesian grammar updates after each concept exposure](#). *Physical Review E*, 101(4):042128.

Peter Worth and Susan Stepney. 2005. [Growing music: Musical interpretations of l-systems](#). In *Applications of Evolutionary Computing*, pages 545–550.

Peter Worth and Susan Stepney. 2021. [Recursive bayesian networks: Generalising and unifying probabilistic context-free grammars and dynamic bayesian networks](#). In *Advances in Neural Information Processing Systems 34*, pages 4370–4383.

Mason Youngblood. 2024. [Language-like efficiency and structure in house finch song](#). *Proceedings of the Royal Society B*, 291(2020):20240250.

George Kingsley Zipf. 2016. *Human behavior and the principle of least effort: An introduction to human ecology*. Ravenio books.

A Tag Reduction for Parse Trees

To reduce the number of node states, we grouped tags with similar functions into the same category. The eleven categories used in our analyses are shown in Table 1.

B Context-free Independence Breaking

In Section 4.4, we presented the results of CFIB when the two nodes are fixed as $(x_0, x_1) = (\text{NP}, \text{NP})$. In this section, we present the results with $(x_0, x_1) = (\text{NP}, \text{VP})$ and (VP, VP) to examine how the metric depends on the fixed nodes. From the results shown in Figure 9, we can observe a power-law decay similar to that with $(x_0, x_1) = (\text{NP}, \text{NP})$, although the power law exponents may be different. This implies that the CFIB in syntactic structures follows a power-law decay, irrespective of the fixed nodes.

C Unbinarized Case

As discussed in Section 3.1, we binarized syntactic trees during preprocessing. To verify that this binarization does not significantly affect our results, we analyze the statistical properties of the unbinarized trees in this section. The MI in sequences is unaffected by binarization. In addition, the dataset is too limited to reliably compute the CFIB for unbinarized trees. Therefore, we focus on the MI in structures and the growth of sequential distance. The results, shown in Figure 10, exhibit the power-law decay of the MI and the slow growth of the sequential distance, which are qualitatively consistent with those observed in the binarized case, although, as expected, the structural distances are smaller. These suggest that the statistical behavior of syntactic structures is robust to binarization.

D WikiText

In the main text, we analyzed the BLLIP corpus. To address whether similar behaviors are observed in other corpora, we perform the same analyses for the WikiText-103 corpus

Tags	Part-of-Speech Tags and Phrase Labels
NP	NP NN NNP NNS CD PRP TMP POS QP NNPS EX NX FW
VP	VP VBD VB VBN VBG VBZ VBP TO
S	S S1 SBAR SINV FRAG SQ SBARQ RRC
PP	IN PP RP PRT
DT	DT PRP\$ PDT
JJ	JJ ADJP JJR JJS NAC ADJ
Others	, . " ' \$: PRN -RRB- -LRB- # X INTJ UH SYM LS LST HYPH NFP
AUX	AUX MD AUXG
RB	RB ADVP RBR RBS
CC	CC UCP CONJP
WH	WHNP WDT WP WHADVP WRB WHPP WP\$ WHADJP NML

Table 1: Eleven tags used in this study and the corresponding part-of-speech tags and phrase labels.

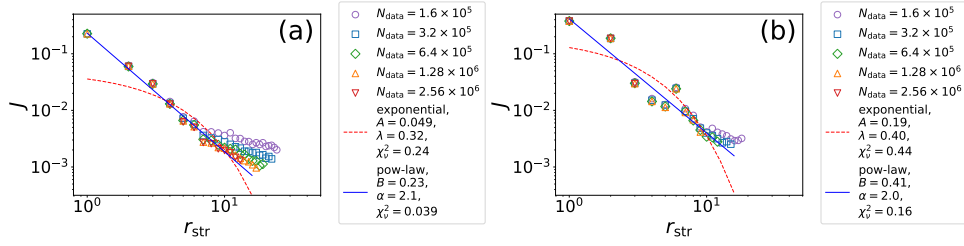


Figure 9: CFIB J for (a) $(x_0, x_1) = (NP, VP)$ and (b) $(x_0, x_1) = (VP, VP)$. Fitting was performed for $N_{data} = 2.56 \times 10^6$ in (a) and $N_{data} = 6.4 \times 10^5$ in (b).

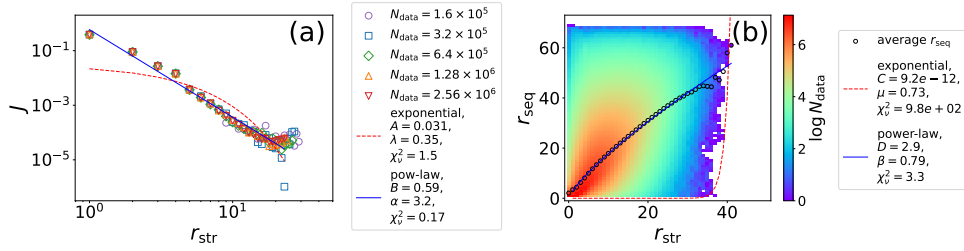


Figure 10: Results for unbinarized parse trees. (a) MI I_{tag} between POS or phrasal tags. Fitting was performed for $N_{data} = 2.56 \times 10^6$. (b) Distribution and average of sequential distances.

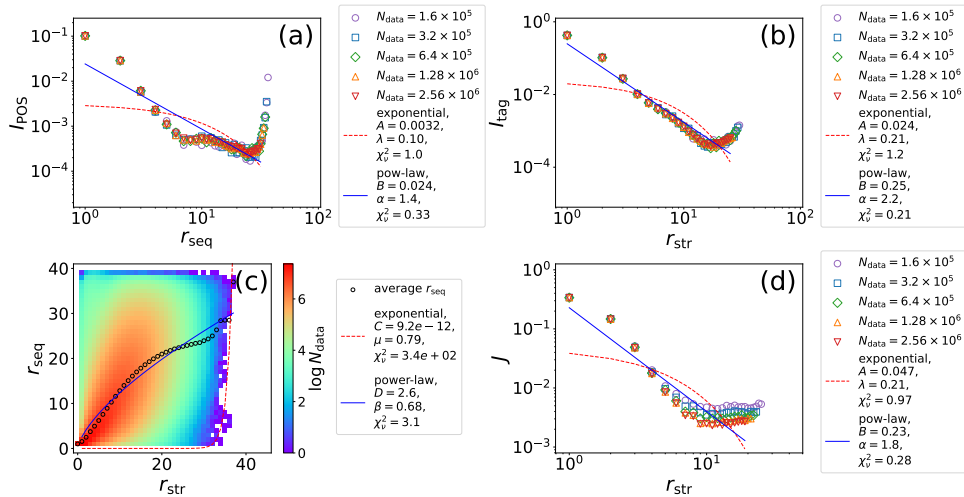


Figure 11: Results for WikiText. (a) MI I_{POS} between POS tags. (b) MI I_{tag} between POS or phrasal tags. (c) Distribution and average of sequential distances. (d) CFIB J for $(x_0, x_1) = (NP, NP)$. Fitting was performed for $N_{data} = 2.56 \times 10^6$ in (a), (b), and (d).

(wikitext-103-raw-v1 split) (Merity et al., 2017), which is a collection of English Wikipedia articles that preserves the original Wikipedia markup and contains over 100 million tokens and approximately 3.9 million sentences. We implement a rule-based detokenizer and segment sentences with a SPACY sentencizer (Honni-bal et al., 2020). Then, we extract sentences up to 40 words (resulting in approximately 3.4 million sentences) and apply the Berkeley Neural Parser (benepar_en3 model) (Kitaev and Klein, 2018; Kitaev et al., 2019) to them to obtain Penn Treebank-style constituency trees. Finally, we binarize the trees and reduce the number of node states using the same method applied to the BLLIP corpus.

Figure 11 shows that the results are qualitatively similar to those for BLLIP: the MI decays according to a power-law function in both sequences and structures; the growth of sequential distance is slower than linear; and the CFIB also decays following a power law. These suggest that the observed behaviors are universal, and that the assumption in Lin and Tegmark (2017) does not hold for syntactic structures across corpora.

However, we note a slight difference in the behavior of MI in both sequences and structures: it increases at large distances, $r_{\text{seq}} \gtrsim 30$ and $r_{\text{str}} \gtrsim 20$. This may be due to structural differences in unusually long sentences or potential parsing errors in such cases. We are unable to determine the cause of this effect, as the dataset contains too few long sentences.